

Investigation of the Continuum Limit in the Presence of Small Breaking of Gauge Invariance in Lattice Gauge Theories

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May 23, 2026

This thesis investigates whether and how the continuum limit of a lattice gauge theory is affected by small explicit violations of gauge invariance. The study combines theoretical scaling arguments with numerical measurements of Wilson loops, Creutz ratios, the static potential, the Sommer scale, and related dimensionless observables. The central goal is to determine whether gauge-breaking effects vanish, persist, or modify the extrapolation toward the continuum limit as the lattice spacing is reduced.

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Notation and Conventions

a Lattice spacing.

β Bare lattice coupling parameter.

ε_1 Bare parameter controlling the explicit breaking of gauge invariance.

$W(R, T)$ Wilson loop with spatial extent R and temporal extent T .

$\chi(R, T)$ Creutz ratio used to estimate the force and string tension.

r_0 Sommer scale defined through $r^2 F(r) = 1.65$.

σ String tension.

1. Introduction

1.1. Context and Problem

Gauge invariance is one of the defining structural principles of lattice gauge theory. In Wilson's formulation it is preserved exactly at nonzero lattice spacing, which strongly constrains the allowed lattice artifacts and supports a controlled approach to the continuum limit. In practical and generalized settings, however, gauge invariance may be only approximate. Small explicit violations can arise from regulator artifacts, effective descriptions, algorithmic imperfections, or imperfect implementations of gauge constraints.

The central problem of this thesis is therefore not whether exact gauge invariance is useful, but how robust the continuum limit is when the lattice theory is perturbed by a controlled, small gauge-breaking deformation. This requires a comparison between the standard gauge-invariant theory and deformed ensembles along a sequence of lattice spacings.

1.2. Research Gap

The theoretical expectation depends on the scaling character of the gauge-breaking deformation. If the deformation acts like an irrelevant lattice artifact, its effect should vanish as the lattice spacing is reduced. If it contains a relevant component, it may change the continuum theory unless it is tuned away. The numerical study must therefore distinguish a harmless change of cutoff effects from a genuine shift of continuum observables.

1.3. Aim and Research Questions

The aim of the thesis is to test this distinction numerically for pure $SU(2)$ lattice gauge theory with a small explicit gauge-breaking deformation. The guiding research question is:

Does a small explicit breaking of gauge invariance disappear in the continuum limit, or does it alter the continuum extrapolation of physical observables?

1.4. Strategy and Main Observables

The numerical strategy is to compare gauge-invariant and gauge-broken ensembles through observables that have a well-defined continuum interpretation. The main observables are temporal Wilson loops, Creutz ratios, the static potential $V(R)$, the Sommer scale r_0 , and the string tension σ or dimensionless combinations derived from it. These quantities provide both scale setting information and direct probes of the continuum extrapolation.

1.5. Structure of the Thesis

Chapter 2 summarizes the theoretical background needed for the analysis. Chapter 3 derives the concrete research gap and formulates the gauge-breaking framework and working hypotheses. Chapter 4 describes the simulation setup, observable extraction, continuum extrapolation, and uncertainty estimation. Chapter 5 presents the numerical results. Chapter 6 interprets the findings in relation to the research question and the theoretical expectations. The final chapter summarizes the conclusions and identifies directions for further research.

2. Scientific Background and Theoretical Framework

The aim of this chapter is to collect the theoretical ingredients needed for the analysis of the continuum limit in the presence of small explicit violations of gauge invariance. The discussion starts from continuum Yang–Mills theory and then introduces the lattice formulation, Wilson loops, the static potential, Creutz ratios, and the Sommer scale. These are the central concepts used later to define and compare physical observables at different lattice spacings.

2.1. Gauge Theories in the Continuum

2.1.1. Gauge Fields and Local Gauge Symmetry

A non-Abelian gauge theory is built from fields that transform locally under a Lie group. In this thesis the relevant case is the group $SU(2)$. A continuum gauge field can be written as

$$A_\mu(x) = A_\mu^a(x)T^a, \quad (2.1)$$

where T^a are the generators of the Lie algebra of $SU(2)$. A common choice is $T^a = \sigma^a/2$, with σ^a the Pauli matrices. The gauge field defines the covariant derivative

$$D_\mu = \partial_\mu + iA_\mu, \quad (2.2)$$

and the associated field-strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]. \quad (2.3)$$

The commutator term distinguishes a non-Abelian gauge theory from an Abelian one. It implies that the gauge field itself carries charge under the gauge group and therefore self-interacts.

A local gauge transformation is specified by a group element

$$\Omega(x) \in SU(2) \quad (2.4)$$

at every point in space-time. Under such a transformation, matter fields in the fundamental representation transform as

$$\psi(x) \mapsto \Omega(x)\psi(x), \quad (2.5)$$

while the gauge field transforms inhomogeneously,

$$A_\mu(x) \mapsto \Omega(x) A_\mu(x) \Omega^\dagger(x) - i(\partial_\mu \Omega(x)) \Omega^\dagger(x). \quad (2.6)$$

The inhomogeneous derivative term is necessary because the transformation is local. The field strength transforms covariantly,

$$F_{\mu\nu}(x) \mapsto \Omega(x) F_{\mu\nu}(x) \Omega^\dagger(x), \quad (2.7)$$

so traces of products of field strengths at the same point are gauge invariant.

In Euclidean space-time, the pure Yang–Mills action can be written as

$$S_{\text{YM}} = \frac{1}{2g^2} \int d^4x \operatorname{tr} F_{\mu\nu}(x) F_{\mu\nu}(x), \quad (2.8)$$

with summation over repeated Lorentz indices. This action is invariant under local gauge transformations. The coupling g controls the strength of the interaction. For an asymptotically free gauge theory, the effective coupling becomes small at short distances, while the long-distance theory is strongly coupled and requires non-perturbative methods.

2.1.2. Gauge-Invariant Observables

Physical observables in a gauge theory must be gauge invariant. Gauge-dependent quantities can be useful after a gauge fixing has been imposed, but they do not directly correspond to physical observables unless the gauge dependence is properly controlled. For pure gauge theory, important gauge-invariant observables include traces of closed parallel transporters, local action-density operators, correlation functions of gauge-invariant operators, and quantities derived from Wilson loops.

A central continuum object is the path-ordered parallel transporter along a curve C ,

$$U(C) = \mathcal{P} \exp \left(i \int_C dx^\mu A_\mu(x) \right). \quad (2.9)$$

For a closed curve, the trace

$$W(C) = \frac{1}{N_c} \operatorname{Re} \operatorname{tr} U(C) \quad (2.10)$$

is gauge invariant. On the lattice this object becomes the Wilson loop and is one of the most direct probes of confinement and of the static quark–antiquark potential.

2.1.3. Physical Consequences of Gauge Invariance

Gauge invariance is not only a formal redundancy of the description. It constrains the allowed terms in the action, the form of correlation functions, and the possible continuum limits. In the lattice formulation, preserving gauge invariance exactly at nonzero lattice spacing is usually advantageous because it prevents gauge-noninvariant counterterms

from being generated by radiative corrections. Wilson’s lattice gauge action realizes this idea by putting compact group-valued variables on links and by constructing the action from traces of closed loops [1, 2, 3].

The purpose of this thesis is to study what happens when this exact symmetry is slightly violated by an explicit deformation. From a renormalization-group point of view, the key question is whether the deformation behaves like an irrelevant lattice artifact, so that its effects vanish as $a \rightarrow 0$, or whether it defines a relevant perturbation that changes the continuum limit. This question is not purely formal: small violations of gauge constraints can appear as regulator artifacts, algorithmic imperfections, or implementation errors in effective or quantum-simulation formulations of lattice gauge theories [4, 5].

2.2. Lattice Gauge Theory

2.2.1. Discretization of Space-Time

In lattice gauge theory, Euclidean space-time is replaced by a hypercubic lattice

$$\Lambda_a = \{x = an \mid n \in \mathbb{Z}^4\}, \quad (2.11)$$

where a is the lattice spacing. The lattice spacing acts as an ultraviolet cutoff. Momenta larger than order $1/a$ are not resolved. The continuum limit is obtained by sending

$$a \rightarrow 0 \quad (2.12)$$

while keeping physical quantities fixed.

In a numerical simulation the lattice has finite extent

$$L_\mu = N_\mu a, \quad (2.13)$$

where N_μ is the number of lattice points in direction μ . Therefore two limits are conceptually distinct: the continuum limit $a \rightarrow 0$ and the infinite-volume limit $L_\mu \rightarrow \infty$. In practice, one tries to choose volumes large enough that finite-volume effects are smaller than the statistical and systematic errors relevant for the analysis. In the simulation campaign of this thesis, spatial volumes are organized to keep the physical spatial extent approximately fixed where possible, while the temporal extent is chosen as part of the ensemble definition and analysis strategy. [Check against the actual simulation setup: in the simulations in this thesis I use $L_\mu = \text{const}$ for the spatial directions and $N_\mu = \text{const}$ for the time direction.]

2.2.2. Link Variables and Plaquettes

The fundamental degrees of freedom of Wilson lattice gauge theory are link variables

$$U_\mu(x) \in \text{SU}(2), \quad (2.14)$$

which live on the oriented link from x to $x + a\hat{\mu}$. They are the lattice analogues of parallel transporters. For a smooth continuum field one may formally write

$$U_\mu(x) \approx \exp \left[iaA_\mu \left(x + \frac{a}{2}\hat{\mu} \right) \right]. \quad (2.15)$$

A local lattice gauge transformation assigns a group element $\Omega(x)$ to each site and acts on the link variables as

$$U_\mu(x) \mapsto \Omega(x)U_\mu(x)\Omega^\dagger(x + a\hat{\mu}). \quad (2.16)$$

The elementary closed loop on the lattice is the plaquette,

$$U_{\mu\nu}(x) = U_\mu(x)U_\nu(x + a\hat{\mu})U_\mu^\dagger(x + a\hat{\nu})U_\nu^\dagger(x). \quad (2.17)$$

Under a gauge transformation, the plaquette transforms covariantly at its base point,

$$U_{\mu\nu}(x) \mapsto \Omega(x)U_{\mu\nu}(x)\Omega^\dagger(x). \quad (2.18)$$

Therefore $\text{Re tr } U_{\mu\nu}(x)$ is gauge invariant.

For smooth fields, the plaquette has the continuum expansion

$$U_{\mu\nu}(x) = \exp \left[ia^2 F_{\mu\nu}(x) + \mathcal{O}(a^3) \right]. \quad (2.19)$$

Thus the plaquette is the lattice object from which the continuum field strength is recovered. Although the later analysis focuses on Wilson loops, static potentials, and Creutz ratios, this relation explains why the Wilson plaquette action reproduces Yang–Mills theory in the continuum limit.

2.2.3. Wilson Gauge Action

The standard Wilson gauge action for $\text{SU}(N_c)$ is

$$S_W[U] = \beta \sum_x \sum_{\mu < \nu} \left(1 - \frac{1}{N_c} \text{Re tr } U_{\mu\nu}(x) \right), \quad (2.20)$$

with

$$\beta = \frac{2N_c}{g_0^2}. \quad (2.21)$$

For $\text{SU}(2)$ this becomes

$$\beta = \frac{4}{g_0^2}, \quad (2.22)$$

and

$$S_W[U] = \beta \sum_x \sum_{\mu < \nu} \left(1 - \frac{1}{2} \text{Re tr } U_{\mu\nu}(x) \right). \quad (2.23)$$

The action is built only from traces of closed loops and is therefore exactly gauge invariant at nonzero lattice spacing.

The continuum limit of the Wilson action is obtained by inserting the plaquette expansion. Up to an irrelevant additive constant and lattice artifacts,

$$S_W[U] \longrightarrow \frac{1}{2g_0^2} \int d^4x \operatorname{tr} F_{\mu\nu}(x) F_{\mu\nu}(x). \quad (2.24)$$

This reproduces the Euclidean Yang–Mills action. The corrections at finite lattice spacing are cutoff effects. For the pure Wilson gauge action, the leading gauge-invariant lattice artifacts in many on-shell quantities are of order a^2 , although the actual scaling observed in numerical data can be affected by statistics, finite-volume effects, operator definitions, and the accessible range of lattice spacings.

The action in Eq. (2.23) is the gauge-invariant reference action. In the numerical study this action is supplemented by a controlled gauge-breaking deformation. The explicit form of this deformation and its symmetry properties are introduced in Chapter 3. The standard Wilson action is kept as the reference theory corresponding to $\epsilon_1 = \epsilon_2 = 0$.

2.2.4. Path Integral and Expectation Values

The lattice path integral is finite-dimensional at finite lattice spacing and finite volume. The partition function is

$$Z = \int \prod_{x,\mu} dU_\mu(x) e^{-S[U]}, \quad (2.25)$$

where $dU_\mu(x)$ is the Haar measure on $SU(2)$. The expectation value of an observable $\mathcal{O}[U]$ is

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \prod_{x,\mu} dU_\mu(x) \mathcal{O}[U] e^{-S[U]}. \quad (2.26)$$

Gauge-invariant expectation values can be computed without gauge fixing. This is one of the practical advantages of the compact Wilson formulation.

In Monte Carlo simulations the path integral is estimated by generating an ensemble of gauge-field configurations distributed according to $e^{-S[U]}$. For an ensemble of N_{cfg} configurations U_i , the estimator is

$$\langle \mathcal{O} \rangle \approx \frac{1}{N_{\text{cfg}}} \sum_{i=1}^{N_{\text{cfg}}} \mathcal{O}[U_i]. \quad (2.27)$$

Because successive configurations in a Markov chain are correlated, autocorrelations must be considered when estimating uncertainties.

2.2.5. Continuum Limit and Renormalization

The continuum limit is not obtained by simply decreasing a at fixed bare coupling. Instead, the bare parameters have to be tuned so that physical observables remain fixed in physical units. For pure $SU(2)$ gauge theory with the Wilson action, decreasing the lattice spacing corresponds to increasing β , or equivalently decreasing the bare coupling g_0 .

A useful way to formulate the continuum limit is in terms of a physical length scale, such as the Sommer scale r_0 . The dimensionless ratio

$$\frac{r_0}{a} \quad (2.28)$$

is measured on the lattice. Increasing r_0/a means that the same physical length is resolved by more lattice points, i.e. the lattice spacing is smaller in physical units. Therefore continuum extrapolations can be expressed as fits in

$$\left(\frac{a}{r_0}\right)^2 \quad (2.29)$$

or another measured dimensionless lattice-spacing proxy.

Continuum extrapolations compare observables at fixed physical conditions, not merely at fixed lattice indices. For observables derived from Wilson loops this means that a loop size R on a coarse ensemble and the same integer R on a fine ensemble do not represent the same physical distance. Therefore one should compare at fixed ratios such as r/r_0 or other fixed dimensionless distances. This point is emphasized in recent work on controlled continuum extrapolations of Wilson-loop observables, where Creutz ratios are used as finite continuum quantities and fitted at fixed physical distance [6].

The renormalization-group interpretation is central for the present thesis. Different lattice actions can lead to the same continuum theory if they differ only by irrelevant operators. In that case they lie in the same universality class. By contrast, a relevant deformation changes the long-distance theory unless its coefficient is tuned to zero. A gauge-breaking term must therefore be classified by its effect on long-distance, gauge-invariant observables as $a \rightarrow 0$. The numerical analysis in later chapters tests this question directly.

2.3. Static Potential and Confinement

2.3.1. Wilson Loops

A Wilson loop is the trace of the ordered product of link variables around a closed contour. For a rectangular loop in the $\mu\nu$ -plane with side lengths L_μ and L_ν in lattice units, the code measures the normalized real trace

$$W_{\mu\nu}(L_\mu, L_\nu) = \frac{1}{N_c V} \sum_x \text{Re tr} \left[U_\mu(x; L_\mu) U_\nu(x + L_\mu \hat{\mu}; L_\nu) U_\mu^\dagger(x + L_\nu \hat{\nu}; L_\mu) U_\nu^\dagger(x; L_\nu) \right], \quad (2.30)$$

where V is the lattice volume and

$$U_\mu(x; L_\mu) = U_\mu(x) U_\mu(x + \hat{\mu}) \cdots U_\mu(x + (L_\mu - 1)\hat{\mu}) \quad (2.31)$$

denotes a straight Wilson line of length L_μ in direction μ . The factor $1/N_c$ normalizes the trace, while the factor $1/V$ averages over all possible base points of the loop.

For temporal Wilson loops, the last lattice direction is used as the Euclidean time direction,

$$t = N_d - 1. \quad (2.32)$$

The code computes rectangular loops in each spatial-time plane and averages over the spatial directions:

$$W(R, T) = \frac{1}{N_d - 1} \sum_{\mu=0}^{N_d-2} W_{\mu,t}(R, T). \quad (2.33)$$

For the SU(2) simulations considered in this thesis, $N_c = 2$, so the stored temporal Wilson loop is

$$W(R, T) = \frac{1}{2V(N_d - 1)} \sum_{\mu=0}^{N_d-2} \sum_x \text{Re tr } W_{\mu,t}(x; R, T). \quad (2.34)$$

Thus the values written to the output file are already normalized by the color factor, averaged over the full lattice volume, and averaged over spatial directions.

In the implementation, the rectangular loop is constructed from two paths that connect opposite corners of the rectangle. The first path is

$$l_\mu = U_\mu(x; L_\mu) U_\nu(x + L_\mu \hat{\mu}; L_\nu), \quad (2.35)$$

and the second path is

$$l_\nu = U_\nu(x; L_\nu) U_\mu(x + L_\nu \hat{\nu}; L_\mu). \quad (2.36)$$

The measured loop contribution is then

$$\text{Re tr } (l_\mu l_\nu^\dagger). \quad (2.37)$$

This is equivalent to the trace of the ordered product around the rectangular closed contour.

The temporal Wilson-loop output file has the structure

```
# step, L, T, W_temp
```

where L is the spatial extent R , T is the temporal extent, and W_temp is the averaged, normalized value defined in Eq. (2.33).

2.3.2. Temporal Smearing and Flow-Time Convention

The signal-to-noise problem for temporal Wilson loops becomes more severe as the loop area increases. The simulations and analysis therefore support two separate smoothing procedures. The first is a multihit estimator for the links entering the temporal Wilson-loop measurement. The second is measurement on a gauge field that has first been evolved by Wilson, or gradient, flow. These two procedures are conceptually distinct: multihit improves the estimator for a given ensemble, while gradient flow changes the gauge field on which the observable is evaluated.

For a temporal loop with spatial extent L and temporal extent T , the measured object can be written as

$$W(L, T) = \frac{1}{N_c} \text{Re tr} \left[S_i(x, L) T_t(x + L\hat{i}, T) S_i^\dagger(x + T\hat{t}, L) T_t^\dagger(x, T) \right], \quad (2.38)$$

followed by the average over all lattice sites x and all spatial directions i . Here $S_i(x, L)$ is the straight spatial transporter of length L in direction i , and $T_t(x, T)$ is the straight temporal transporter of length T . In the implementation the temporal direction is the last lattice direction, $t = N_d - 1$.

The raw measurement constructs the required spatial transporters up to the largest requested spatial length. The temporal transporter is then built iteratively,

$$T_t(x, T + 1) = T_t(x, T) U_t(x + T\hat{t}). \quad (2.39)$$

For each requested pair (L, T) , the closed loop in Eq. (2.38) is accumulated over all origins and spatial orientations. The value written to `W_temp.out` is normalized by

$$(N_d - 1) V N_c, \quad (2.40)$$

where V is the lattice volume. This is the same normalization used in Eq. (2.33).

When multihit temporal smearing is enabled, links entering the loop estimator are replaced by local averages generated with the same local action used for the simulation update. For Metropolis simulations, the local average is estimated from repeated Metropolis proposals for the selected link at fixed surrounding staple. For heatbath simulations, the averaged link is generated from local heatbath updates, optionally followed by the configured number of overrelaxation steps. Setting

$$\text{wilson_loop_multihit} = 1 \quad (2.41)$$

disables this link averaging and gives the raw Wilson-loop estimator.

Gradient flow is handled separately. For each measured configuration U , a copy $V(t)$ is evolved by Wilson flow using a third-order Runge–Kutta integrator. The flow force is constructed from the local staple, and after each flow step the links are projected back onto the gauge group. Wilson loops at positive flow time are measured on the flowed links $V_\mu(x, t)$, not on the original links $U_\mu(x)$.

The flow time stored in the output is the dimensionless lattice flow time

$$t_{\text{lat}} = \frac{t}{a^2}. \quad (2.42)$$

Accordingly, the smoothing radius in lattice units is

$$\frac{\sqrt{8t}}{a} = \sqrt{8t_{\text{lat}}}. \quad (2.43)$$

The gradient-flow output files therefore label flow time as `t_over_a2`. Temporal Wilson loops measured during flow are written to `gradient_flow_wtemp.dat` with the column structure

conf_id, t_over_a2, L, T, W_temp

where `conf_id` is the saved-configuration identifier and `W_temp` is the normalized temporal Wilson loop at the specified flow time.

At flow time $t = 0$, the configured multihit estimator is still used. For positive flow time, the code sets the Wilson-loop multihit count to one, so the smoothing of the measured observable comes only from the gradient flow. This convention avoids combining a flowed observable with an additional stochastic link-averaging procedure.

2.3.3. Effective Potential Extraction

Temporal Wilson loops are used to extract the static quark–antiquark potential. For large Euclidean time extent T , the Wilson loop has the spectral form

$$W(R, T) = \sum_n c_n(R) e^{-aE_n(R)T}, \quad (2.44)$$

where $E_0(R) = V(R)$ is the static ground-state potential and the higher $E_n(R)$ describe excited states with the same static sources. At sufficiently large T , the ground state dominates:

$$W(R, T) \propto e^{-aV(R)T}. \quad (2.45)$$

A lattice effective potential is therefore defined by the ratio

$$aV_{\text{eff}}(R, T) = -\ln \left[\frac{W(R, T+1)}{W(R, T)} \right]. \quad (2.46)$$

If excited-state contributions are negligible, this approaches the desired static potential in lattice units:

$$aV_{\text{eff}}(R, T) \xrightarrow{T \rightarrow \infty} aV(R). \quad (2.47)$$

In the numerical analysis, $W(R, T)$ denotes the value stored as `W_temp`. This quantity is already normalized by the color factor $1/N_c$, averaged over lattice base points, and averaged over the spatial directions used for temporal Wilson loops. For the SU(2) simulations considered here, $N_c = 2$. Therefore no further normalization by color, volume, or spatial directions is applied in the analysis.

2.3.4. Creutz Ratios

Creutz ratios are combinations of neighboring Wilson loops designed to cancel some perimeter and normalization contributions. In the continuum they can be viewed as the mixed derivative

$$\chi(r, t) = -\frac{\partial^2}{\partial r \partial t} \ln W(r, t), \quad (2.48)$$

where $W(r, t)$ is a rectangular Wilson loop of physical spatial extent r and Euclidean time extent t . A central finite-difference discretization gives a lattice Creutz ratio located

at half-integer separations,

$$\chi\left(R + \frac{1}{2}, T + \frac{1}{2}\right) = \ln \left[\frac{W(R+1, T)W(R, T+1)}{W(R, T)W(R+1, T+1)} \right], \quad (2.49)$$

or, equivalently after a shift of indices,

$$\chi(R, T) = -\ln \left[\frac{W(R, T)W(R-1, T-1)}{W(R, T-1)W(R-1, T)} \right]. \quad (2.50)$$

The normalization of the Wilson loop, including the factor $1/N_c$, cancels in this ratio. This is useful for the present $SU(2)$ analysis because the same formula can be applied directly to the normalized Wilson-loop output described above.

The central-difference form approximates the continuum mixed derivative with cutoff effects of order a^2 when the underlying Wilson loop is smooth. This makes Creutz ratios particularly useful for continuum studies: they are finite observables, their leading normalization and perimeter contributions cancel, and they are closely related to the static force. In the large-time limit,

$$\frac{\chi(R, T)}{a^2} \xrightarrow{T \rightarrow \infty} F(r) \quad (2.51)$$

up to the usual distinction between the discrete lattice midpoint and the physical distance r . Diagonal ratios $\chi(R, R)$ are also useful summary observables because they probe the same spatial and temporal scale and are straightforward to compare across ensembles [7, 6].

If large Wilson loops obey an area law,

$$W(R, T) \sim \exp[-\sigma a^2 RT - \text{perimeter terms}], \quad (2.52)$$

then the Creutz ratio approaches

$$\chi(R, T) \xrightarrow{R, T \rightarrow \infty} \sigma a^2. \quad (2.53)$$

Thus Creutz ratios provide a direct lattice estimator for the string tension in lattice units.

At finite R and T , Creutz ratios contain corrections from excited states, Coulombic contributions, finite-volume effects, and lattice artifacts. They are therefore best interpreted either as approximate string-tension estimators at large loop sizes or as diagnostic observables for scaling comparisons between ensembles.

For continuum extrapolations the important distinction is between fixed lattice separation and fixed physical distance. Holding R fixed while taking $a \rightarrow 0$ drives the physical distance $r = Ra$ to zero and therefore probes the short-distance limit rather than the same physical force. The continuum comparison should instead use a dimensionless distance such as

$$\hat{r} = \frac{r}{r_0} \quad (2.54)$$

and interpolate the measured Creutz ratios to common $\hat{r}$ values before fitting in a/r_0 . This is also where gauge-breaking effects are expected to be most visible: short-distance Wilson-loop observables are sensitive to ultraviolet deformations, while large loops suffer from rapidly increasing statistical noise.

2.3.5. Sommer Scale and String Tension

The static potential can also be used to set the scale. A common phenomenological form is the Cornell-type potential

$$V(r) = V_0 - \frac{\alpha}{r} + \sigma r, \quad (2.55)$$

where V_0 is an additive self-energy, α parametrizes the short-distance Coulombic part, and σ is the string tension. The corresponding force is

$$F(r) = \frac{dV}{dr}. \quad (2.56)$$

The additive constant V_0 drops out of the force.

The Sommer scale r_0 is defined implicitly by

$$r_0^2 F(r_0) = 1.65. \quad (2.57)$$

This definition uses the force at intermediate distances, where the potential can often be determined more accurately than at very large distances. In lattice units, the force is estimated from finite differences of the potential. If $\hat{V}(R) = aV(Ra)$, then

$$\hat{F}\left(R + \frac{1}{2}\right) = \hat{V}(R+1) - \hat{V}(R) \quad (2.58)$$

is a dimensionless estimate of $a^2 F(r)$ at an intermediate distance. The Sommer condition can then be written as

$$\left(\frac{r}{a}\right)^2 \hat{F}\left(\frac{r}{a}\right) = 1.65. \quad (2.59)$$

Solving this equation gives r_0/a .

The ratio r_0/a is one of the main scale-setting quantities in this thesis. It allows data at different β values to be compared in approximately physical units and provides the continuum variable

$$\frac{a}{r_0} = \left(\frac{r_0}{a}\right)^{-1}. \quad (2.60)$$

The continuum limit corresponds to

$$\frac{a}{r_0} \rightarrow 0. \quad (2.61)$$

The string tension may also be expressed in dimensionless form as

$$\sigma r_0^2. \quad (2.62)$$

This quantity is useful because it is independent of the arbitrary choice of lattice units. Agreement of such dimensionless quantities along a sequence of lattice spacings is a central test of scaling toward a common continuum limit.

3. Research Gap and Gauge-Breaking Framework

3.1. Research Gap Derived from the Background

The background chapter establishes two facts that are central for this thesis. First, the Wilson plaquette action preserves local gauge invariance exactly at nonzero lattice spacing. Second, the continuum limit is not tested directly on the bare link variables, but through dimensionless observables such as Wilson loops, Creutz ratios, the static potential, the Sommer scale, and combinations involving the string tension. The research gap is the relation between these two facts when the lattice action is deliberately deformed by terms that are not locally gauge invariant.

The standard universality argument says that many changes of the lattice action do not change the continuum theory. If the added terms correspond only to irrelevant cutoff effects, their contribution to physical observables should disappear as $a \rightarrow 0$. Gauge invariance, however, is a particularly restrictive symmetry. Once it is broken at finite lattice spacing, the usual symmetry argument that forbids gauge-noninvariant counterterms no longer applies. A small breaking term can therefore be harmless only if its effect is dynamically suppressed along the continuum trajectory, or if the coefficient is tuned in such a way that the gauge-invariant fixed point is recovered.

This leaves an empirical question for the present simulations. The deformation is known explicitly at the level of the update action, and its strength can be varied. What is not known a priori is whether the measured gauge-invariant observables approach the same continuum values as in the undeformed theory, or whether the deformation shifts the continuum extrapolation. The role of the numerical analysis is therefore not merely to observe a dependence on the breaking parameters at fixed lattice spacing. Such a dependence is expected. The central question is whether this dependence remains in the continuum intercept after the data are expressed in physical units.

3.2. Definition of the Gauge-Breaking Deformation

The simulations start from the Wilson plaquette action and add an explicit single-link term. Up to additive constants that do not affect normalized expectation values, the action used for the Metropolis ensembles is

$$S[U] = -\frac{\beta}{N_c} \sum_p \text{Re tr } U_p - \varepsilon_1 \sum_{x,\mu} \frac{1}{N_c} \text{Re tr } U_\mu(x). \quad (3.1)$$

Here U_p denotes the plaquette matrix, $N_c = 2$ for the SU(2) ensembles studied in this work, and ε_1 is the bare parameter controlling the strength of the explicit breaking. For $N_c = 2$, the linear breaking term is equivalently

$$-\frac{\varepsilon_1}{2} \sum_{x,\mu} \text{Re tr } U_\mu(x). \quad (3.2)$$

The first term is the Wilson plaquette action written without the irrelevant constant. Indeed,

$$\beta \sum_p \left(1 - \frac{1}{N_c} \text{Re tr } U_p\right) = \beta N_p - \frac{\beta}{N_c} \sum_p \text{Re tr } U_p, \quad (3.3)$$

where N_p is the number of plaquettes. The constant βN_p cancels from normalized expectation values. For $\varepsilon_1 = 0$, the action therefore reduces to the standard gauge-invariant Wilson theory.

The second term is not built from closed loops. It acts directly on individual links and is therefore sensitive to the representative chosen on each gauge orbit. Positive ε_1 favours links with large positive $\text{Re tr } U_\mu(x)$, and therefore biases links toward the identity element.

For a proposed replacement of a single link U_{old} by U_{new} , the Metropolis acceptance step uses the local action difference

$$\begin{aligned} \Delta S = & -\frac{\beta}{N_c} [\text{Re tr } (U_{\text{new}} S) - \text{Re tr } (U_{\text{old}} S)] \\ & - \varepsilon_1 \left[\frac{1}{N_c} \text{Re tr } U_{\text{new}} - \frac{1}{N_c} \text{Re tr } U_{\text{old}} \right], \end{aligned} \quad (3.4)$$

where S is the usual staple sum coupled to the updated link. This is the local form of Eq. (3.1).

The symmetry breaking follows directly from the lattice gauge transformation law in Eq. (2.16). The Wilson plaquette term is invariant because each plaquette transforms by conjugation at its base point. A single link transforms as

$$U_\mu(x) \mapsto \Omega(x) U_\mu(x) \Omega^\dagger(x + \hat{\mu}), \quad (3.5)$$

and therefore

$$\text{Re tr } U_\mu(x) \mapsto \text{Re tr } [\Omega(x) U_\mu(x) \Omega^\dagger(x + \hat{\mu})]. \quad (3.6)$$

For independent transformations at the two endpoints of the link, this is not equal to the original trace in general. Thus nonzero ε_1 explicitly breaks the local gauge symmetry. The exact local redundancy of the Wilson theory is replaced by a theory whose Boltzmann weight distinguishes between different representatives of the same Wilson gauge orbit.

One exact symmetry remains. If the same group element is used at every lattice site,

$$\Omega(x) = \Omega, \quad (3.7)$$

then each link transforms by conjugation,

$$U_\mu(x) \mapsto \Omega U_\mu(x) \Omega^\dagger, \quad (3.8)$$

and the trace is unchanged. The deformed action therefore preserves the global diagonal $SU(N_c)$ subgroup while breaking the independent site-dependent gauge transformations.

The deformation acts as an explicit preference for particular representatives on each gauge orbit. It is not a gauge fixing of the Wilson theory: the Boltzmann weight itself is changed. The central question of this thesis is whether this explicit gauge-orbit bias disappears from gauge-invariant observables as the continuum limit is approached.

3.3. Symmetry and Scaling Expectations

3.3.1. Relevant, Marginal, and Irrelevant Breaking Terms

The continuum interpretation of Eq. (3.1) depends on how the breaking terms scale near the gauge-invariant continuum limit. In renormalization-group language there are three qualitatively different possibilities. If the deformation is irrelevant, it changes the finite-spacing approach to the continuum but not the limiting theory. If it is marginal, its effect can remain visible over a broad range of lattice spacings and may require a more detailed analysis of logarithmic or weak scaling violations. If it contains a relevant component, then a fixed nonzero breaking strength changes the continuum theory unless the corresponding bare coefficient is tuned to zero.

The single-link trace terms are local and ultra-violet in character. They act on individual links rather than on extended physical observables. This makes it plausible that small values of ε_1 may first appear as cutoff distortions in short-distance quantities such as small Wilson loops and short Creutz ratios. Plausibility, however, is not a proof of irrelevance. Since the terms are not protected by local gauge symmetry, the numerical analysis must allow both outcomes: a shared continuum intercept with modified cutoff effects, or a shifted continuum intercept.

3.3.2. Possible Restoration of Gauge Invariance

Gauge invariance can be effectively restored in the continuum limit if the breaking contribution to physical observables vanishes as $a \rightarrow 0$. In that case the deformed ensembles and the Wilson ensembles lie on different lattice trajectories but approach the same continuum fixed point. For a dimensionless observable $\hat{O}$, this situation has the schematic form

$$\hat{O}(\hat{a}, \varepsilon_1, \dots) = O_0 + c_a \hat{a}^2 + c_1 \varepsilon_1 \hat{a}^{p_1} + \dots, \quad p_1, p_2 > 0, \quad (3.9)$$

where $\hat{a} = a/r_0$. The coefficients and powers are observable dependent, but the important point is the common intercept O_0 .

If restoration does not occur at fixed breaking strength, the leading dependence can instead enter the continuum intercept,

$$\hat{O}(\hat{a}, \varepsilon_1) = O_0 + d_1 \varepsilon_1 + c_a \hat{a}^2 + \dots. \quad (3.10)$$

In this case the deformed theory approaches a different continuum limit, or no controlled common continuum limit is visible in the accessible range. The distinction between Eqs. (3.9) and (3.10) is the main target of the continuum fits in Chapter 4.

3.3.3. Expected Dependence on Breaking Strength and Lattice Spacing

At fixed β and fixed volume, increasing ε_1 should increase the average link trace because the action gives lower weight to links far from the identity. This also affects Wilson loops indirectly, since products of links become biased toward larger traces. For sufficiently small ε_1 , the leading response of a generic observable is expected to be approximately linear unless a symmetry forbids the linear term. In the Metropolis implementation it is therefore even under $\text{Re tr } U_\mu(x) \mapsto -\text{Re tr } U_\mu(x)$ at the level of a single link, but it is still gauge breaking and can shift gauge-invariant observables through the generated ensemble distribution.

The lattice-spacing dependence is the decisive part of the test. If the breaking terms only modify cutoff effects, then differences between broken and unbroken ensembles should shrink when observables are compared at common physical distance and plotted against $\hat{a}^2$. If the deformation changes the continuum theory, the broken data may remain separated from the gauge-invariant baseline as $\hat{a} \rightarrow 0$. The methodology therefore uses two classes of fits: shared-intercept fits, which encode restoration of gauge invariance, and separate-intercept fits, which test for a persistent breaking effect.

The analysis also uses the rescaled parameter

$$\bar{\varepsilon} = \frac{\varepsilon_1}{\beta} \left(\frac{r_0}{a} \right)^2, \quad (3.11)$$

as a diagnostic for comparing ensembles at different lattice spacings. This quantity is not a substitute for the bare action parameters in Eq. (3.1); rather, it provides a dimensionless way to organize the observed breaking strength relative to the measured scale.

3.4. Working Hypotheses

The numerical analysis tests the following hypotheses.

1. For sufficiently small ε_1 , gauge-invariant observable can approach the same continuum values as in the Wilson theory, even though local gauge symmetry is broken at finite lattice spacing.
2. At fixed lattice spacing, the deformation produces measurable shifts in short-distance observables, especially the average link trace, small Wilson loops, and small Creutz ratios.

3. If the breaking behaves as an irrelevant cutoff effect, the shifts between broken and unbroken ensembles decrease with a positive power of a/r_0 after comparison at fixed physical distance.
4. If the deformation contains a relevant component, fixed nonzero ε_1 produces continuum extrapolations whose intercepts are incompatible with the gauge-invariant baseline.
5. There may be a practical small-breaking regime. Beyond it, the response to ε_1 can become nonlinear, continuum fits can become unstable, and the assumptions behind a perturbative breaking analysis no longer apply.

4. Methodological Approach

4.1. Simulation Setup and Parameters

The gauge-field ensembles were generated for pure SU(2) lattice gauge theory on four-dimensional hypercubic lattices. Each ensemble is specified by an `input.yaml` file, which contains the lattice geometry $N_0 \times N_1 \times N_2 \times N_3$, the bare coupling β , the gauge-breaking parameters ε_1 , the Metropolis update parameters, and the measurement setup. The simulations use the Wilson gauge action supplemented by the gauge-breaking deformation introduced in Chapter 3. Gauge configurations are generated with a local Metropolis algorithm. One sweep consists of updating all link variables of the lattice, with `nHits` Metropolis proposals per link. The proposal size is controlled by the parameter `delta`. The main bare parameters varied in the simulation campaign are the coupling β , which controls the lattice spacing, and the breaking strength ε_1 . Measurements are performed after fixed sweep intervals and include the plaquette, the average link trace, and temporal Wilson loops. The temporal Wilson loops are measured for selected spatial and temporal extents (R, T) and are written to `W_temp.out`; these data form the basis for the extraction of the static potential, Creutz ratios, and scale setting quantities.

4.2. Ensemble Organization and Fixed-Physics Logic

Explain how ensembles are grouped by common physical and algorithmic parameters. State which parameters are varied in the continuum study and which are held fixed. The organizing principle is that comparisons across β should be made at fixed physical conditions as far as the available data allow, rather than at fixed lattice indices alone.

4.3. Thermalization, Autocorrelations, and Bootstrap

4.3.1. Thermalization Cuts

Describe how the initial non-equilibrated part of each Markov chain is identified and removed before combining runs.

4.3.2. Integrated Autocorrelation Time

Estimate or monitor autocorrelation times for representative observables in order to choose measurement spacing and bootstrap block lengths.

4.3.3. Bootstrap Blocking

The measurements after thermalization are still not statistically independent, because consecutive configurations are generated by a Markov chain. A naive bootstrap over individual measurements would therefore underestimate the statistical error. To account for this, the ordered measurement history is divided into contiguous blocks of fixed length b in simulation time. The bootstrap samples are then generated by drawing these blocks with replacement and averaging over the resampled history. Unless stated otherwise, the statistical errors quoted in this work use $N_{\text{boot}} = 1000$ bootstrap replicas.

The block length must be large compared with the autocorrelation time of the observables used in the analysis. If b is too small, correlations remain between the resampled data and the error estimate is biased downward. If b is too large, the number of available blocks becomes small and the bootstrap estimate becomes noisy. In practice, several block sizes are compared and the chosen value is taken from the region where the estimated errors are stable under further increases of b .

The same bootstrap samples are propagated through the nonlinear analysis steps. Wilson loops are resampled first, and derived quantities such as effective potentials, the static potential $V(R)$, Creutz ratios, and the Sommer scale are recomputed for each bootstrap replica. This preserves correlations between quantities derived from the same ensemble and gives the statistical uncertainties used in the later continuum analysis.

4.4. Analysis Pipeline and Data Reduction

The final analysis starts from the individual simulation runs and groups together runs with the same base parameters, in particular the same β , lattice size, ε_1 . Runs in such a group differ only by statistical details such as the random seed or the total number of sweeps and are treated as independent Markov chains for the same ensemble.

For each input run the early Monte Carlo history is inspected before combining the data. In practice, Wilson loops and simple observables are plotted as a function of the simulation step. A thermalization cutoff is chosen from these plots. Measurements before this cutoff are discarded, and the remaining measurements from all runs in the group are combined into one data set.

The combined post-thermalization data are then used to estimate the Wilson loops $W(R, T)$ and their statistical uncertainties. Errors are obtained by a blocked bootstrap, with a block size chosen large enough to reduce the effect of autocorrelations. From the Wilson loops, effective masses are formed and fit windows in T are selected to extract the static potential $V(R)$.

The finalized potential points are fitted with a Cornell-type ansatz to obtain the Sommer scale r_0/a . This scale is then used to form the dimensionless lattice spacing

$$\hat{a} = \frac{a}{r_0} \tag{4.1}$$

and the rescaled breaking parameter

$$\bar{\varepsilon} = \frac{\varepsilon_1}{\beta} \left(\frac{r_0}{a} \right)^2. \quad (4.2)$$

Creutz ratios and available gradient-flow observables are evaluated from the same combined ensemble. The resulting finalized ensemble summaries are the inputs for the later comparison across different β values and breaking strengths.

4.5. Extraction of Physical Observables

4.5.1. Wilson-Loop Analysis

Describe the effective-mass plateaus, the choice of temporal fit windows, and the systematic uncertainty associated with varying those windows.

4.5.2. Static Potential

Describe the potential extraction, Cornell-type fits, force determination, and Sommer-scale extraction.

4.5.3. Creutz-Ratio Analysis

The finalized analysis computes Creutz ratios directly from the measured temporal Wilson loops, not from the fitted static potential. The input data are the post-thermalization `W_temp` measurements of the individual simulation runs belonging to one ensemble. For each run, the analysis reads `W_temp.out`. If gradient-flow Wilson loops were produced, the additional Wilson-loop file named in `input.yaml` can also be loaded and stored under its nonzero flow-time labels, but the default finalized $R = T$ Creutz-ratio plot uses the unflowed $t_{\text{flow}} = 0$ Wilson loops unless the analysis explicitly requests another flow time. Each row contains the simulation step, the spatial loop size L , the temporal loop size T , and the normalized temporal Wilson-loop value `W_temp`. After applying the chosen thermalization cut for each run, the remaining measurements with the same (L, T) are concatenated across all runs in the ensemble. This combined Wilson-loop history is the only data source used for the standard Creutz ratios in the finalized output.

For every available adjacent set of loop sizes, the code forms

$$\chi \left(R + \frac{1}{2}, T + \frac{1}{2} \right) = \ln \left[\frac{W(R, T+1)W(R+1, T)}{W(R, T)W(R+1, T+1)} \right]. \quad (4.3)$$

Thus a displayed diagonal point at $R = T = 1.5$, for example, is calculated from the four measured Wilson loops $W(1, 1)$, $W(1, 2)$, $W(2, 1)$, and $W(2, 2)$. More generally, the diagonal plot `creutz_ratios_R_eq_T.html` keeps only those Creutz-ratio midpoints for which the spatial and temporal midpoint are equal. The horizontal coordinate is therefore the midpoint value $R + 1/2 = T + 1/2$, not an original integer loop extent.

The central values are computed by first averaging each of the four required Wilson-loop histories and then inserting these four means into the logarithmic ratio. Statistical errors are obtained by the same blocked bootstrap used for the Wilson-loop analysis. In each bootstrap replica, contiguous blocks of saved configurations are resampled, the four Wilson-loop means are recomputed from the resampled histories, and the Creutz ratio is recalculated from those bootstrap means. The standard deviation of the resulting bootstrap distribution is stored as the Creutz-ratio error. Bootstrap samples for each accepted ratio are also written to the finalized analysis directory.

In the finalized output, the complete set of ratios is saved in `creutz/summary.json` under the keys `chi`, `chi_err`, and `chi_bootstrap_paths`. The keys of `chi` are midpoint pairs such as `1.5,1.5`. The summary script used for scans then reads this file and extracts the positive-within-error diagonal entries as columns named `chi_R=T_1.5`, `chi_R=T_2.5`, and so on. These summary columns are therefore derived from the finalized Creutz-ratio JSON file, which in turn is derived from the combined post-thermalization `W_temp` Wilson-loop measurements of the source runs.

4.5.4. Scale Setting

Summarize the determination of the Sommer scale in lattice units, possible alternative scale estimates, and consistency checks.

4.6. Continuum Extrapolation and Scaling Tests

4.6.1. Choice of Continuum Variable

The primary continuum variable is chosen to be the measured dimensionless lattice spacing

$$\hat{a} = \frac{a}{r_0}. \quad (4.4)$$

For observables that are expected to have leading even-power cutoff effects with the Wilson gauge action, the default horizontal axis is therefore $\hat{a}^2 = (a/r_0)^2$. Using the measured scale rather than the bare coupling makes the extrapolation closer to a fixed-physics comparison and reduces the risk of interpreting changes in scale setting as physical changes in the observable itself.

For Creutz-ratio observables the quantity to extrapolate should be dimensionless. Convenient choices are

$$\hat{\chi}(\hat{r}) = r_0^2 \frac{\chi(r,t)}{a^2}, \quad \hat{r} = \frac{r}{r_0}, \quad (4.5)$$

or, for large loops where the string-tension interpretation is appropriate, σr_0^2 . The measured lattice ratio $\chi(R,T)$ is dimensionless and approaches σa^2 in the area-law regime; division by a^2 and multiplication by r_0^2 converts it to a quantity with a finite physical continuum limit. The same logic applies to the force extracted from the static potential.

4.6.2. Fixed-Physics Trajectories

Ensembles at different β values are compared along trajectories where the physical volume and the dimensionless analysis point are kept as fixed as the available data allow. In practice this means comparing observables at common $\hat{r} = r/r_0$ and using only ensembles whose volumes are large enough that finite-volume corrections are not expected to dominate the Wilson-loop signal. When exact matching is not possible, the mismatch is included as a systematic effect or avoided by restricting the fit range.

The gauge-breaking parameters are treated as additional bare deformation parameters. If they are held fixed while a changes, the continuum fit tests whether their effect vanishes dynamically or survives in the continuum intercept. If a theoretical scaling prescription for ε is used, for example $\varepsilon \propto a^q$, the fit must instead include this scaling explicitly. The central diagnostic is whether the broken and unbroken data approach a common continuum value.

4.6.3. Fit Ansatz

The fit strategy compares a gauge-invariant baseline with combined lattice-spacing and breaking-dependence models.

Gauge-invariant baseline. For the gauge-invariant ensembles, a minimal continuum ansatz for a dimensionless observable $\hat{O}$ is

$$\hat{O}(\hat{a}) = O_0 + c_a \hat{a}^2 + c_{a4} \hat{a}^4 + \dots \quad (4.6)$$

The $\hat{a}^4$ term is included only when the data support it or when it is used as a fit-systematic variation. The default fit should be stable under removing the coarsest lattice spacing and under modest changes of the physical-distance range used for the observable.

Combined lattice-spacing and breaking dependence. A useful shared-intercept test is

$$\hat{O}(\hat{a}, \varepsilon) = O_0 + c_a \hat{a}^2 + c_\varepsilon \varepsilon \hat{a}^p + c_{a\varepsilon} \varepsilon \hat{a}^{p+2} + \dots, \quad (4.7)$$

with $p > 0$. This ansatz encodes the hypothesis that gauge breaking is an irrelevant cutoff effect: for any fixed small ε the breaking terms vanish as $\hat{a} \rightarrow 0$, and all data share the same continuum intercept O_0 .

The competing separate-intercept test is

$$\hat{O}(\hat{a}, \varepsilon) = O_0 + c_\varepsilon \varepsilon + c_a \hat{a}^2 + c_{a\varepsilon} \varepsilon \hat{a}^2 + \dots \quad (4.8)$$

Here a statistically significant c_ε indicates that the breaking deformation changes the continuum value of the observable rather than merely changing cutoff effects. If the deformation can enter symmetrically in the sign of ε , a quadratic term $c_{\varepsilon^2} \varepsilon^2$ should be included as a systematic variant.

This structure follows the same logic used in controlled continuum extrapolations with smearing or flow: one separates changes in the path toward the continuum from changes in the continuum observable itself [6]. In that setting, fixed smearing in lattice units

can share the same continuum intercept if it remains in the controlled expansion regime, while a fixed physical smoothing radius defines a different continuum observable. In the present gauge-breaking problem there is no guarantee of a shared intercept, so Eqs. (4.7) and (4.8) are both part of the analysis.

4.6.4. Uncertainty Budget

Statistical errors. Statistical errors are obtained from blocked bootstrap samples. The block size is chosen large enough to reduce residual autocorrelation effects in the saved Markov-chain history. Derived observables such as Creutz ratios, Sommer scales, and continuum-fit parameters are recomputed inside the bootstrap whenever possible, so nonlinear error propagation is captured by the resampling.

Fit-window and interpolation systematics. Wilson-loop observables depend on analysis choices: the temporal range used for effective-potential plateaus, the loop sizes included in Creutz-ratio summaries, and the interpolation used to compare fixed physical distances. These choices are varied to estimate systematic effects. For Creutz ratios, short distances are most vulnerable to cutoff and breaking artifacts, while long distances are most vulnerable to statistical noise.

Finite-volume and continuum-fit systematics. Finite-volume effects are monitored by comparing ensembles with similar $\hat{a}$ and ε but different physical volumes when available. Continuum-fit systematics are estimated by changing the fit ansatz, removing the coarsest lattice spacing, and checking whether the extrapolation remains monotonic over the fitted range. Non-monotonic behavior or strong shifts of the intercept under reasonable fit variations are treated as evidence that the available lattice spacings are not yet in a controlled scaling regime.

4.7. Computational Environment

The simulations were carried out on the QBig cluster. Most production runs were performed on nodes 13–17 equipped with NVIDIA P100 GPUs. Additional and faster runs were performed on nodes 18 and 19 equipped with NVIDIA A100 GPUs. The GPU type affects the wall-clock runtime but not the generated Markov-chain ensemble, provided that the same action, random seed handling, and update algorithm are used.

5. Results

5.1. Thermalization and Data Quality

Insert representative thermalization histories and autocorrelation estimates.

5.2. Gauge-Invariant Baseline

Present the $\varepsilon = 0$ results for $W(R, T)$, $V(R)$, r_0 , and σ . This establishes the reference continuum behavior.

5.3. Dependence on Gauge-Breaking Strength

5.3.1. Observable Shifts at Fixed Coupling

Report how each observable changes with ε_1 at fixed β and fixed lattice geometry.

5.3.2. Scaling of Shifts with Breaking Strength

Fit or summarize whether the shifts are linear, quadratic, or nonlinear in the breaking strength within the accessible range.

5.3.3. Thresholds or Nonlinear Behavior

Identify any breaking strengths for which the data no longer follow the small-deformation pattern assumed in the scaling analysis.

5.4. Continuum Limit at Fixed Breaking

5.4.1. Continuum Extrapolations for Each Breaking Strength

Present continuum fits separately for each investigated breaking strength.

5.4.2. Comparison with the Gauge-Invariant Limit

Compare the extrapolated values with the $\varepsilon = 0$ baseline and state whether the intercepts are statistically compatible.

5.4.3. Stability Under Fit Variations

Show how the continuum conclusions change when fit ranges, lattice spacings, and systematic variants are modified.

5.5. Combined Scaling Analysis

Fit the dependence on both lattice spacing and gauge-breaking parameter. State whether the data support restoration of gauge invariance in the continuum limit.

5.6. Finite-Volume and Systematic Checks

Collect checks that are not part of the primary result but constrain the interpretation.

6. Discussion

6.1. Interpretation of the Continuum Limit

Discuss whether small gauge breaking behaves like an irrelevant lattice artifact, a relevant deformation, or a source of uncontrolled continuum bias.

6.2. Comparison with Theoretical Expectations

Relate the numerical findings to the hypotheses from Chapter [3](#).

6.3. Limitations

Discuss limits from statistics, accessible lattice spacings, finite volume, fit-systematic control, and the specific form of gauge breaking studied.

6.4. Implications

Explain what the result means for simulations or formulations where gauge invariance is only approximate.

7. Conclusion and Outlook

7.1. Summary of Findings

State the central quantitative conclusions.

7.2. Answer to the Research Question

Give the direct answer to whether the continuum limit is robust under the investigated small breaking of gauge invariance.

7.3. Future Work

Possible extensions:

- Study additional forms of gauge-breaking terms.
- Increase statistics and extend to finer lattice spacings.
- Add gauge-variant diagnostics to directly measure symmetry restoration.
- Compare different gauge groups or dimensions.

A. Simulation Tables

Provide complete tables of ensembles, including β , lattice size, breaking parameters, number of sweeps, thermalization cuts, and measured scales.

B. Additional Plots

Include supplementary Wilson-loop plateaus, bootstrap block-size scans, potential fits, and continuum-fit variants.

C. Analysis Code and Reproducibility

Document the exact command-line workflow, cache version, software environment, and data-selection rules used for the final analysis.

D. Derivations

Collect longer derivations that would interrupt the main text, such as Creutz-ratio identities or continuum-scaling arguments.

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